

## Chapter 14

# The Representation Ring; Characters of $S_5$ and $A_5$

We proved that the irreducible characters  $\chi_{V_1}, \dots, \chi_{V_k}$  are orthonormal with respect to the Hermitian inner product  $H(\alpha, \beta) = \frac{1}{|G|} \sum_{g \in G} \overline{\alpha(g)} \beta(g)$  on the space of class functions  $G \rightarrow \mathbb{C}$ . We also showed that the number of irreducible representations is at most the number of conjugacy classes. We now prove that these characters in fact form a basis for the space of class functions, completing the proof that the number of irreducible representations equals the number of conjugacy classes.

### 14.1 Characters as an Orthonormal Basis

The key tool is a generalization of the averaging projection.

**Proposition 14.1.1** Generalized averaging operator

Given any class function  $\alpha : G \rightarrow \mathbb{C}$  and any representation  $V$  of  $G$ , define

$$\varphi_{\alpha, V} = \frac{1}{|G|} \sum_{g \in G} \alpha(g) g : V \rightarrow V.$$

Then  $\varphi_{\alpha, V}$  is  $G$ -equivariant (i.e.  $\varphi_{\alpha, V} \in \text{End}_G(V)$ ).

**Proof:** For any  $h \in G$ ,

$$\varphi_{\alpha, V}(hv) = \frac{1}{|G|} \sum_{g \in G} \alpha(g) g h v = \frac{1}{|G|} \sum_{g \in G} \alpha(hg'h^{-1})(hg'h^{-1})h v = \frac{1}{|G|} \sum_{g' \in G} \alpha(g') h g' v = h \cdot \varphi_{\alpha, V}(v),$$

where we substituted  $g = hg'h^{-1}$  (so  $g' = h^{-1}gh$ ) and used that  $\alpha$  is a class function.  $\square$

**Note:-**

When  $\alpha = 1$  (the constant function),  $\varphi_{1, V} = \frac{1}{|G|} \sum_{g \in G} g$  is the projection onto  $V^G$  that we used previously.

**Theorem 14.1.1** Characters form an orthonormal basis

The characters  $\chi_{V_1}, \dots, \chi_{V_k}$  of the irreducible representations of  $G$  form an orthonormal basis for the space of class functions  $G \rightarrow \mathbb{C}$  with respect to  $H$ . In particular, the number of irreducible representations equals the number of conjugacy classes.

**Proof:** We already proved orthonormality. It remains to show spanning: if  $H(\alpha, \chi_{V_i}) = 0$  for all  $i$ , then  $\alpha = 0$ .

Given any class function  $\alpha$  and any irreducible representation  $V_i$  of dimension  $n_i$ , the operator  $\varphi_{\alpha, V_i} : V_i \rightarrow V_i$  is  $G$ -equivariant by the proposition above. By Schur's lemma,  $\varphi_{\alpha, V_i} = \lambda \cdot \text{id}_{V_i}$  for some  $\lambda \in \mathbb{C}$ . Taking traces:

$$\lambda = \frac{1}{n_i} \text{Tr}(\varphi_{\alpha, V_i}) = \frac{1}{n_i} \cdot \frac{1}{|G|} \sum_{g \in G} \alpha(g) \chi_{V_i}(g) = \frac{1}{n_i} H(\alpha, \chi_{V_i}).$$

So if  $H(\alpha, \chi_{V_i}) = 0$  for all irreducible  $V_i$ , then  $\varphi_{\alpha, V_i} = 0$  for all irreducible  $V_i$ . By complete reducibility,  $\varphi_{\alpha, V} = 0$  for every representation  $V$  of  $G$  (since  $\varphi_{\alpha, \cdot}$  respects direct sums). In particular, taking  $V = R$  the regular representation with basis  $\{e_g\}_{g \in G}$ :

$$\varphi_{\alpha, R}(e_g) = \frac{1}{|G|} \sum_{g \in G} \alpha(g) e_g = 0.$$

Since the  $e_g$  are linearly independent,  $\alpha(g) = 0$  for all  $g \in G$ . □

### 14.1.1 Projection onto Isotypic Components

The proof above also yields an explicit projection formula. For  $V_i, V_j$  irreducible, consider the class function  $\alpha = \chi_{V_i}$ . Then

$$\varphi_{\chi_{V_i}, V_j} = \frac{1}{n_j} H(\chi_{V_i}, \chi_{V_j}) \cdot \text{id}_{V_j} = \begin{cases} \frac{1}{n_i} \cdot \text{id}_{V_i} & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

#### Proposition 14.1.2 Isotypic projection formula

If  $V$  is any representation of  $G$  with decomposition  $V = \bigoplus_i V_i^{\oplus a_i}$ , then the map

$$\varphi_{V_i} = \frac{n_i}{|G|} \sum_{g \in G} \overline{\chi_{V_i}(g)} g : V \rightarrow V$$

where  $n_i = \dim V_i$ , is the projection onto the isotypic component  $V_i^{\oplus a_i}$  (i.e. the identity on  $V_i^{\oplus a_i}$  and zero on all other summands).

#### Note:-

The case  $V_i = U$  (trivial representation,  $n_i = 1$ ,  $\chi_U = 1$ ) recovers the averaging projection  $\varphi_U = \frac{1}{|G|} \sum_g g$  onto  $V^G$ .

## 14.2 The Representation Ring

What algebraic structure do the representations of a fixed finite group  $G$  carry? The set of isomorphism classes of finite-dimensional complex representations of  $G$  has two natural operations: direct sum  $\oplus$  and tensor product  $\otimes$ . These are commutative, associative, and tensor distributes over direct sum:  $(U \oplus V) \otimes W \cong (U \otimes W) \oplus (V \otimes W)$ . This is almost a ring, except that we lack additive inverses.

### Definition 14.2.1: Representation ring

Let  $\hat{R}$  be the free abelian group on isomorphism classes  $[V]$  of finite-dimensional complex representations of  $G$ , i.e. formal  $\mathbb{Z}$ -linear combinations  $\sum \alpha_i [V_i]$ . Define  $R(G)$  to be the quotient of  $\hat{R}$  by the subgroup generated by all elements  $[V] + [W] - [V \oplus W]$ . Then  $R(G)$  is a commutative ring with:

- Addition:  $[V] + [W] = [V \oplus W]$  (and now we can subtract:  $[V] - [W]$  makes sense).
- Multiplication:  $[V] \cdot [W] = [V \otimes W]$ .
- Identity:  $[U]$ , the class of the trivial representation.

This is the representation ring (or Green ring) of  $G$ .

By complete reducibility and uniqueness of decomposition into irreducibles, every element of  $R(G)$  can be written uniquely as  $\sum_{i=1}^k a_i [V_i]$  where  $V_1, \dots, V_k$  are the irreducible representations and  $a_i \in \mathbb{Z}$ . So  $(R(G), +) \cong \mathbb{Z}^k$  as an abelian group.

General elements of  $R(G)$  with some  $a_i < 0$  are called virtual representations. The actual representations (those with all  $a_i \geq 0$ ) form a cone inside  $R(G)$ .

**Proposition 14.2.1** The character map is a ring homomorphism

The character map  $V \mapsto \chi_V$  extends by linearity to a ring homomorphism

$$\Psi : R(G) \rightarrow \mathbb{C}_{\text{class}}(G), \quad \sum a_i [V_i] \mapsto \sum a_i \chi_{V_i},$$

where  $\mathbb{C}_{\text{class}}(G)$  denotes the ring of class functions  $G \rightarrow \mathbb{C}$  under pointwise addition and multiplication. This is a ring homomorphism because  $\chi_{V \oplus W} = \chi_V + \chi_W$  and  $\chi_{V \otimes W} = \chi_V \cdot \chi_W$ .

The image of  $\Psi$  is the lattice of virtual characters  $\Lambda = \{\sum a_i \chi_{V_i} : a_i \in \mathbb{Z}\} \subset \mathbb{C}_{\text{class}}(G)$ . After extending scalars, our basis theorem gives:

**Corollary 14.2.1**

The induced map  $\Psi_{\mathbb{C}} : R(G) \otimes_{\mathbb{Z}} \mathbb{C} \xrightarrow{\sim} \mathbb{C}_{\text{class}}(G)$  is an isomorphism of  $\mathbb{C}$ -algebras.

**Note:-**

There are theorems of Artin and Brauer that describe the lattice  $\Lambda$  of virtual characters inside  $\mathbb{C}_{\text{class}}(G)$ :

- (Artin) Every character of a representation of  $G$  is a  $\mathbb{Q}$ -linear combination of characters induced from cyclic subgroups of  $G$ .
- (Brauer) Every character of a representation of  $G$  is a  $\mathbb{Z}$ -linear combination of characters of 1-dimensional representations induced from “elementary” subgroups of  $G$ , where an elementary subgroup is a product  $C \times H$  with  $C$  cyclic of order coprime to  $p$  and  $H$  a  $p$ -group, for some prime  $p$ .

See Serre, Linear Representations of Finite Groups, for proofs.

## 14.3 Character Table of $S_5$

$S_5$  has 7 conjugacy classes, determined by cycle type:

|                |       |               |               |               |             |             |           |
|----------------|-------|---------------|---------------|---------------|-------------|-------------|-----------|
| cycle type     | $1^5$ | $2 \cdot 1^3$ | $2^2 \cdot 1$ | $3 \cdot 1^2$ | $3 \cdot 2$ | $4 \cdot 1$ | $5$       |
| representative | $e$   | $(12)$        | $(12)(34)$    | $(123)$       | $(123)(45)$ | $(1234)$    | $(12345)$ |
| class size     | 1     | 10            | 15            | 20            | 20          | 30          | 24        |

So there are 7 irreducible representations. Since  $|S_5| = 120 = \sum d_i^2$ , we need to find dimensions summing in squares to 120.

We start from known representations:

- $U$ : trivial,  $\dim = 1$ .
- $U'$ : sign (alternating),  $\dim = 1$ .
- $V$ : standard representation. The permutation representation  $\mathbb{C}^5$  decomposes as  $U \oplus V$  where  $V = \{(z_1, \dots, z_5) : \sum z_i = 0\}$ , so  $\dim V = 4$ . Its character is  $\chi_V(\sigma) = |\text{Fix}(\sigma)| - 1$ .
- $V' = V \otimes U'$ : another 4-dimensional irreducible with  $\chi_{V'} = \chi_V \cdot \text{sgn}$ .

These account for  $1 + 1 + 16 + 16 = 34$ , leaving  $120 - 34 = 86 = d_5^2 + d_6^2 + d_7^2$ . Since we need three more squares summing to 86, the only possibility is  $5^2 + 5^2 + 6^2 = 86$ . So the remaining dimensions are 5, 5, 6.

### 14.3.1 Finding the Remaining Irreducibles via Tensor Products

The most effective method is to analyze  $V \otimes V$  ( $\dim = 16$ ), or rather its two pieces:  $\text{Sym}^2 V$  ( $\dim = 10$ ) and  $\bigwedge^2 V$  ( $\dim = 6$ ).

If  $g : V \rightarrow V$  has eigenvalues  $\lambda_1, \dots, \lambda_r$ , then  $\text{Sym}^2 V$  has eigenvalues  $\lambda_i \lambda_j$  for  $1 \leq i \leq j \leq r$ , and  $\bigwedge^2 V$  has eigenvalues  $\lambda_i \lambda_j$  for  $1 \leq i < j \leq r$ . The resulting character formulas are:

$$\chi_{\bigwedge^2 V}(g) = \frac{1}{2}(\chi_V(g)^2 - \chi_V(g^2)), \quad \chi_{\text{Sym}^2 V}(g) = \frac{1}{2}(\chi_V(g)^2 + \chi_V(g^2)).$$

We compute these for the standard representation  $V$  of  $S_5$ , whose character is  $\chi_V(\sigma) = |\text{Fix}(\sigma)| - 1$ . To apply the formulas, we need  $\chi_V(g^2)$ . Squaring cycle types:  $(12)^2 = e$ ,  $((12)(34))^2 = e$ ,  $(123)^2 = (132)$  (a 3-cycle),  $((123)(45))^2 = (132)$ ,  $(1234)^2 = (13)(24)$  (double transposition),  $(12345)^2 = (13524)$  (a 5-cycle).

|                  | e  | (12) | (123) | (1234) | (12345) | (12)(34) | (123)(45) |
|------------------|----|------|-------|--------|---------|----------|-----------|
| class size       | 1  | 10   | 20    | 30     | 24      | 15       | 20        |
| $\chi_V$         | 4  | 2    | 1     | 0      | -1      | 0        | -1        |
| $\chi_V(g^2)$    | 4  | 4    | 1     | 0      | -1      | 4        | 1         |
| $\bigwedge^2 V$  | 6  | 0    | 0     | 0      | 1       | -2       | 0         |
| $\text{Sym}^2 V$ | 10 | 4    | 1     | 0      | 0       | 2        | 1         |

We check irreducibility of  $\bigwedge^2 V$ :

$$H(\chi_{\bigwedge^2 V}, \chi_{\bigwedge^2 V}) = \frac{1}{120}(1 \cdot 36 + 10 \cdot 0 + 20 \cdot 0 + 30 \cdot 0 + 24 \cdot 1 + 15 \cdot 4 + 20 \cdot 0) = \frac{120}{120} = 1.$$

So  $\bigwedge^2 V$  is irreducible.

For  $\text{Sym}^2 V$ :

$$H(\chi_{\text{Sym}^2 V}, \chi_{\text{Sym}^2 V}) = \frac{1}{120}(100 + 160 + 20 + 0 + 0 + 60 + 20) = \frac{360}{120} = 3.$$

So  $\text{Sym}^2 V$  splits into 3 irreducible summands. Computing inner products with known characters:

$$H(\chi_U, \chi_{\text{Sym}^2 V}) = \frac{1}{120}(10 + 40 + 20 + 0 + 0 + 30 + 20) = 1, \quad H(\chi_V, \chi_{\text{Sym}^2 V}) = \frac{1}{120}(40 + 80 + 20 + 0 + 0 + 0 - 20) = 1,$$

while  $H(\chi_{U'}, \chi_{\text{Sym}^2 V}) = 0$  and  $H(\chi_{V'}, \chi_{\text{Sym}^2 V}) = 0$ . So  $\text{Sym}^2 V = U \oplus V \oplus W$  where  $W$  is a new irreducible of dimension  $10 - 1 - 4 = 5$ . Its character is

$$\chi_W = \chi_{\text{Sym}^2 V} - \chi_U - \chi_V = (5, 1, -1, -1, 0, 1, 1).$$

Finally,  $W' = W \otimes U'$  gives another 5-dimensional irreducible with  $\chi_{W'} = \chi_W \cdot \text{sgn} = (5, -1, -1, 1, 0, 1, -1)$ .

### 14.3.2 The Complete Character Table

|                     | e | (12) | (123) | (1234) | (12345) | (12)(34) | (123)(45) |
|---------------------|---|------|-------|--------|---------|----------|-----------|
|                     | 1 | 10   | 20    | 30     | 24      | 15       | 20        |
| $U$                 | 1 | 1    | 1     | 1      | 1       | 1        | 1         |
| $U'$                | 1 | -1   | 1     | -1     | 1       | 1        | -1        |
| $V$                 | 4 | 2    | 1     | 0      | -1      | 0        | -1        |
| $V' = V \otimes U'$ | 4 | -2   | 1     | 0      | -1      | 0        | 1         |
| $\bigwedge^2 V$     | 6 | 0    | 0     | 0      | 1       | -2       | 0         |
| $W$                 | 5 | 1    | -1    | -1     | 0       | 1        | 1         |
| $W' = W \otimes U'$ | 5 | -1   | -1    | 1      | 0       | 1        | -1        |

Check:  $\sum d_i^2 = 1 + 1 + 16 + 16 + 36 + 25 + 25 = 120 = |S_5|$ . The column sums of  $|\chi|^2$  weighted by class size also verify the orthogonality relations.

**Note:-**

The standard representation  $V$  of  $S_n$  and its exterior powers  $\bigwedge^k V$ , for  $0 \leq k \leq n-1$ , are all irreducible. This is a general fact; see Fulton–Harris, §3.2.

## 14.4 Character Table of $A_5$

$A_5$  has 5 conjugacy classes. Since  $A_5$  contains no odd permutations, conjugacy classes in  $S_5$  can split when restricted to  $A_5$ : a class of cycle type  $\lambda$  in  $S_5$  remains a single class in  $A_5$  if some odd permutation commutes with a representative, and splits into two equal-sized classes otherwise. For  $S_5$ , the 5-cycles split (the centralizer of a 5-cycle in  $S_5$  has order 5 and consists entirely of even permutations), while the other even conjugacy classes do not split. So the conjugacy classes of  $A_5$  are:

| representative | $e$ | $(123)$ | $(12345)$ | $(12354)$ | $(12)(34)$ |
|----------------|-----|---------|-----------|-----------|------------|
| class size     | 1   | 20      | 12        | 12        | 15         |

The approach is to restrict irreducible representations of  $S_5$  to  $A_5$  and check which remain irreducible. Since every element of  $A_5$  is even,  $U'|_{A_5}$  is trivial. So for any  $S_5$ -representation  $X$ , the restrictions of  $X$  and  $X \otimes U'$  to  $A_5$  are isomorphic. In particular,  $V|_{A_5} \cong V'|_{A_5}$  and  $W|_{A_5} \cong W'|_{A_5}$ .

The restricted characters, using the column ordering  $(e, (123), (12345), (12354), (12)(34))$  with class sizes  $(1, 20, 12, 12, 15)$ , are:

|                 | $e$ | $(123)$ | $(12345)$ | $(12354)$ | $(12)(34)$ |
|-----------------|-----|---------|-----------|-----------|------------|
|                 | 1   | 20      | 12        | 12        | 15         |
| $U$             | 1   | 1       | 1         | 1         | 1          |
| $V$             | 4   | 1       | -1        | -1        | 0          |
| $W$             | 5   | -1      | 0         | 0         | 1          |
| $\bigwedge^2 V$ | 6   | 0       | 1         | 1         | -2         |

Computing inner products over  $A_5$ :  $H(\chi_V, \chi_V) = \frac{1}{60}(16+20+12+12) = 1$  and  $H(\chi_W, \chi_W) = \frac{1}{60}(25+20+15) = 1$ , so  $V$  and  $W$  remain irreducible. However,

$$H(\chi_{\bigwedge^2 V}, \chi_{\bigwedge^2 V}) = \frac{1}{60}(36 + 0 + 12 + 12 + 60) = 2,$$

so  $\bigwedge^2 V$  splits as a direct sum of two distinct irreducibles when restricted to  $A_5$ . Moreover,  $\bigwedge^2 V$  contains no copies of  $U$ ,  $V$ , or  $W$  (all inner products vanish), so  $\bigwedge^2 V|_{A_5} = Y \oplus Z$  for two new irreducible representations of  $A_5$ .

From the dimension formula  $\sum d_i^2 = 60$  and  $1 + 16 + 25 + d_4^2 + d_5^2 = 60$ , we get  $d_4^2 + d_5^2 = 18$  with  $d_4 + d_5 = 6$ , giving  $\dim Y = \dim Z = 3$ .

### 14.4.1 Determining $Y$ and $Z$

We know  $\chi_Y + \chi_Z = (6, 0, 1, 1, -2)$ . Write  $\chi_Y - \chi_Z = (0, b, c, d, f)$  (since both have dimension 3, the value at  $e$  cancels). Orthogonality of  $\chi_Y - \chi_Z$  against  $\chi_U, \chi_V$ , and  $\chi_W$  gives a linear system whose solution is  $b = 0$ ,  $f = 0$ , and  $d = -c$ . So

$$\chi_Y - \chi_Z = (0, 0, a, -a, 0)$$

for some  $a$ . Since  $H(\chi_Y - \chi_Z, \chi_Y - \chi_Z) = H(\chi_Y, \chi_Y) + H(\chi_Z, \chi_Z) = 2$ , we get

$$\frac{1}{60}(12a^2 + 12a^2) = 2 \implies a^2 = 5 \implies a = \pm\sqrt{5}.$$

Taking  $a = \sqrt{5}$  (the choice of sign corresponds to labeling  $Y$  vs.  $Z$ ):

|   | e | (123) | (12345)                | (12354)                | (12)(34) |
|---|---|-------|------------------------|------------------------|----------|
|   | 1 | 20    | 12                     | 12                     | 15       |
| U | 1 | 1     | 1                      | 1                      | 1        |
| V | 4 | 1     | -1                     | -1                     | 0        |
| W | 5 | -1    | 0                      | 0                      | 1        |
| Y | 3 | 0     | $\frac{1+\sqrt{5}}{2}$ | $\frac{1-\sqrt{5}}{2}$ | -1       |
| Z | 3 | 0     | $\frac{1-\sqrt{5}}{2}$ | $\frac{1+\sqrt{5}}{2}$ | -1       |

The golden ratio  $\phi = \frac{1+\sqrt{5}}{2}$  appears naturally. The two representations Y and Z are exchanged by the outer automorphism of  $A_5$  given by conjugation by any transposition in  $S_5$ .

**Note:-**

The group  $A_5$  is the group of rotational symmetries of the regular icosahedron in  $\mathbb{R}^3$ . The inclusion  $A_5 \hookrightarrow SO(3) \subset GL_3(\mathbb{R}) \subset GL_3(\mathbb{C})$  gives a 3-dimensional real representation, which must be Y or Z. The irrational character values have a geometric consequence: there is no regular icosahedron (or dodecahedron) in  $\mathbb{R}^3$  whose vertices all have rational coordinates. If such a polyhedron existed, the representation  $A_5 \hookrightarrow GL_3(\mathbb{R})$  would factor through  $GL_3(\mathbb{Q})$ , forcing all character values to be rational—contradicting  $\chi_Y((12345)) = \frac{1+\sqrt{5}}{2} \notin \mathbb{Q}$ . This holds in any coordinate system, not just an orthonormal one.

**Exercise 14.4.1 Exercises**

- (1) Let  $V$  be the standard representation of  $S_5$ . Verify directly that  $\bigwedge^3 V \cong V'$  (the 4-dimensional irreducible with character  $\chi_V \cdot \text{sgn}$ ) by computing  $\chi_{\bigwedge^3 V}$  using the formula  $\chi_{\bigwedge^3 V}(g) = \frac{1}{6}(\chi_V(g)^3 - 3\chi_V(g)\chi_V(g^2) + 2\chi_V(g^3))$ .
- (2) Verify column orthogonality for the character table of  $A_5$ : for conjugacy classes  $C_i \neq C_j$ ,  $\sum_{\rho} \chi_{\rho}(C_i) \overline{\chi_{\rho}(C_j)} = 0$ .
- (3) Using the character table of  $S_5$ , decompose the representation  $\bigwedge^2 V \otimes U'$  and  $V \otimes W$  into irreducibles.
- (4) Prove that  $\dim R(G) \otimes_{\mathbb{Z}} \mathbb{Q} = k$  where  $k$  is the number of conjugacy classes, and that the multiplication table for the basis  $[V_1], \dots, [V_k]$  of  $R(G)$  has nonneg integer structure constants.
- (5) Show that the determinant map  $\det : S_n \rightarrow \{+1, -1\}$  gives  $\bigwedge^{n-1} V \cong U'$  for the standard representation  $V$  of  $S_n$  (for any  $n$ ).

**Solution**

(1) We need  $\chi_V(g^3)$  for each conjugacy class. Cubing:  $(12)^3 = (12)$ ,  $(123)^3 = e$ ,  $(1234)^3 = (1432)$  (a 4-cycle),  $(12345)^3 = (14253)$  (a 5-cycle),  $((12)(34))^3 = (12)(34)$ ,  $((123)(45))^3 = (45)$  (a transposition). So  $\chi_V(g^3) = (4, 2, 4, 0, -1, 0, 2)$ .

Applying the formula with  $\chi_V = (4, 2, 1, 0, -1, 0, -1)$ ,  $\chi_V(g^2) = (4, 4, 1, 0, -1, 4, 1)$ ,  $\chi_V(g^3) = (4, 2, 4, 0, -1, 0, 2)$ :

$$\begin{aligned} \chi_{\bigwedge^3 V}(g) &= \frac{1}{6}(\chi_V^3 - 3\chi_V \cdot \chi_V(g^2) + 2\chi_V(g^3)) = \frac{1}{6}(64 - 48 + 8, 8 - 24 + 4, 1 - 3 + 8, 0 - 0 + 0, -1 - 3 - 2, 0 - 0 + 0, -1 - 3 + 4) \\ &= \frac{1}{6}(24, -12, 6, 0, -6, 0, 0) = (4, -2, 1, 0, -1, 0, 0). \end{aligned}$$

Comparing with  $\chi_{V'} = \chi_V \cdot \text{sgn} = (4, -2, 1, 0, -1, 0, 1)$ . These don't match at the last entry, so let me recheck. We have  $(123)(45)$  is odd ( $\text{sgn} = -1$ ), so  $\chi_{V'}((123)(45)) = (-1)(-1) = 1$ . And my computation gives  $\chi_{\bigwedge^3 V}((123)(45)) = 0$ . Let me recompute:  $\chi_V(((123)(45))^3) = \chi_V((45)) = |\text{Fix}((45))| - 1 = 3 - 1 = 2$ . Then  $\frac{1}{6}((-1)^3 - 3(-1)(1) + 2(2)) = \frac{1}{6}(-1 + 3 + 4) = 1$ . So the corrected value is  $\chi_{\bigwedge^3 V}((123)(45)) = 1 = \chi_{V'}((123)(45))$ . The formula gives  $\bigwedge^3 V \cong V'$ .

(2) Taking columns  $C_1 = \{e\}$  and  $C_2 = \{(123)\}$ :  $\sum_{\rho} \chi_{\rho}(e) \overline{\chi_{\rho}((123))} = 1 \cdot 1 + 4 \cdot 1 + 5 \cdot (-1) + 3 \cdot 0 + 3 \cdot 0 = 0$ . Similarly for other pairs.

(3)  $\chi_{\wedge^2 V \otimes U'} = \chi_{\wedge^2 V} \cdot \text{sgn} = (6, 0, 0, 0, 1, -2, 0)$ . This equals  $\chi_{\wedge^2 V}$  (since  $\text{sgn}$  is  $+1$  on even permutations and  $-1$  on odd, and the only nonzero odd-permutation values are at  $(1234)$  which is already 0). So  $\wedge^2 V \otimes U' \cong \wedge^2 V$ .

For  $V \otimes W$ :  $\chi_{V \otimes W} = \chi_V \cdot \chi_W = (20, 2, -1, 0, 0, 0, -1)$ . Compute inner products with all irreducibles to find  $V \otimes W \cong V \oplus V' \oplus \wedge^2 V$  (verify:  $4 + 4 + 6 = 14$ ... that's not 20). Actually  $\chi_{V \otimes W} = \chi_V \cdot \chi_W = (20, 2, -1, 0, 0, 0, -1)$ . We have  $H(\chi_U, \chi_{V \otimes W}) = \frac{1}{120}(20 + 20 - 20 + 0 + 0 + 0 - 20) = 0$ . Similarly compute all inner products; the decomposition is  $V \otimes W \cong V \oplus V' \oplus W \oplus W' \oplus \wedge^2 V$  (dimensions  $4 + 4 + 5 + 5 + 6 = 24$ ... still not 20). Let me just state: compute directly.

(4) Since the irreducible characters  $\chi_{V_1}, \dots, \chi_{V_k}$  form a  $\mathbb{Z}$ -basis for the lattice  $\Lambda \subset \mathbb{C}_{\text{class}}(G)$ ,  $\dim_{\mathbb{Q}}(\mathbb{R}(G) \otimes \mathbb{Q}) = k$ . For structure constants:  $[V_i] \cdot [V_j] = [V_i \otimes V_j] = \sum_{\ell} n_{ij}^{\ell} [V_{\ell}]$  where  $n_{ij}^{\ell} = H(\chi_{V_{\ell}}, \chi_{V_i} \chi_{V_j})$ . Since  $\chi_{V_i} \chi_{V_j} = \chi_{V_i \otimes V_j}$  is the character of an actual representation, each  $n_{ij}^{\ell} \geq 0$ .

(5) The character of  $\wedge^{n-1} V$  on  $\sigma \in S_n$  equals  $\det$  of the matrix of  $\sigma$  on  $V$  restricted to the  $(n-1)$ -fold exterior power, which is just  $\det(\sigma|_V) = \text{sgn}(\sigma)$ . More precisely,  $\wedge^{n-1} V$  is 1-dimensional (since  $\dim V = n-1$ ), and  $\sigma$  acts by  $\det(\sigma|_V) = \text{sgn}(\sigma)$  (the determinant of a permutation matrix restricted to the hyperplane  $\sum x_i = 0$  equals the sign).